

Customer Segmentation & Churn Pattern Analytics in European Banking

Submitted by

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Chapter 1. Abstract

Customer churn is a significant analytical and strategic challenge for retail banking because the loss of existing customers can reduce customer lifetime value, increase acquisition requirements, and create instability in the customer base. This study analyses customer churn patterns in a European banking dataset containing 10,000 customer records. The analysis focuses on identifying high-risk customer segments, comparing churn across geographic and demographic groups, and assessing the financial characteristics of churned customers.

A structured segmentation framework was developed using geography, age, credit score, tenure, and balance. The analysis found an overall churn rate of 20.37%, with Germany showing the highest geographic churn rate at 32.44%. Customers aged 46–60 recorded the highest age-segment churn rate at 51.12%, while the combination of Germany and the 46–60 age group produced the highest observed segment-level churn rate at 67.33%. Female customers had a higher churn rate than male customers, at 25.07% compared with 16.46%. High-balance customers also showed elevated churn, with a 25.23% churn rate.

The financial analysis identified 1,211 high-balance customers among the churned population, representing 85.94% of the total balance exposure associated with churned customers. The relationship between salary and balance among high-balance churners was negligible, with a Pearson correlation coefficient of 0.0123. These findings indicate that churn is concentrated in a particular geographic, age, engagement, and financial segments rather than being evenly distributed across the customer base. The results provide an evidence-based foundation for targeted retention strategies and are further operationalised through an interactive Streamlit dashboard.

Chapter 2. Introduction

2.1 Background

Customer retention is an important consideration in retail banking because customers interact with financial products and services over extended periods and may generate value through deposits, product usage, transactions, and other banking relationships. When customers leave a bank, the organisation may face the combined challenge of losing an existing relationship and having to invest additional resources to acquire replacement customers. For this reason, understanding which customer groups experience higher levels of churn is important for effective customer management.

Modern banking datasets provide detailed customer-level information that can be used to examine differences in churn behaviour. Variables such as geography, age, credit score, tenure, account balance, product holdings, credit-card ownership, activity status, and estimated salary allow customer populations to be segmented and compared.

This project therefore adopts a structured customer segmentation approach to examine churn patterns in a European banking context. The analysis considers geographic, demographic, engagement, tenure, and financial characteristics, with particular attention given to high-value customers and the balance exposure associated with churn.

2.2 Problem Statement

Despite having rich customer-level data, banks face challenges in identifying high-risk customer segments, understanding churn differences by geography and demographics, and quantifying the financial profile of churned customers. Without structured analytics, churn management strategies may remain generic, reactive, and inefficient.

2.3 Project Objectives

1. Measure the overall customer churn rate.
2. Identify churn distribution across customer segments.
3. Compare churn behaviour across European regions.
4. The project also addresses the following secondary objectives:
5. Understand churn among high-value customers.
6. Evaluate engagement and tenure patterns associated with churn.
7. Support strategic planning and marketing decisions.

2.4 Scope of the Study

The study uses customer-level banking data covering 10,000 records. The analytical scope is centred on descriptive and comparative churn analysis rather than predictive modelling. Customer segments are constructed using predefined rules for geography, age, credit score, tenure, and balance. The study examines churn rates, churn contribution, customer profiles, engagement patterns, and financial exposure among churned customers. The results are subsequently presented through an interactive dashboard to support practical interpretation.

Chapter 3. Literature Review

Customer churn analysis has become an important area of business analytics because organisations need to understand not only how many customers leave, but also whether churn is concentrated among customer groups. In banking, this is especially relevant because customer relationships can involve multiple products and relatively long customer lifecycles. A segmentation-based approach can therefore provide more actionable information than an aggregate churn measure alone.

3.1 Customer Churn in Banking

Banking churn is commonly examined as a customer retention problem in which the analytical objective is to distinguish customers who remain with an institution from those who terminate or otherwise exit their relationship. Previous banking churn research has frequently used customer demographics, account characteristics, product usage, and behavioural indicators to identify patterns associated with customer departure. Such studies demonstrate the usefulness of customer-level data for understanding heterogeneous churn behaviour.

3.2 Customer Segmentation and Churn Analysis

Customer segmentation provides a structured way of comparing groups that may have different levels of churn. Geographic segmentation can reveal differences between markets, while demographic segmentation can identify variation across age and gender groups. Financial

segmentation can further distinguish customers according to measures such as account balance or credit-related characteristics. Combining these dimensions can reveal interactions that are not visible when each variable is considered independently.

For this project, segmentation is deliberately designed around interpretable business categories. Geography is represented by France, Germany, and Spain. Age is divided into Under 30, 30–45, 46–60, and 60+. Credit score is grouped into Low, Medium, and High. Tenure is categorised as New, Mid-term, and Long-term, while balance is classified as Zero Balance, Low Balance, and High Balance. This structure supports direct comparison of churn rates and customer composition across meaningful groups.

3.3 Demographic and Geographic Factors

Demographic characteristics can be associated with differences in customer behaviour, financial needs, and engagement with banking services. Age, for example, may correspond with differences in financial circumstances and product requirements, while gender and geography may capture differences in customer composition or market conditions. However, observed differences should be interpreted as associations rather than evidence that a demographic characteristic directly causes churn.

3.4 Engagement, Tenure and Product Characteristics

Customer engagement is another important dimension of retention analysis. Indicators such as active-member status and product holdings can provide insight into the strength of a customer's relationship with a bank. Tenure can also be examined to determine whether churn is concentrated among newer customers or whether longer-standing customers show comparable risk. These variables are particularly useful when combined with demographic and financial segmentation.

3.5 Financial Characteristics and High-Value Customers

Financial characteristics are relevant because the departure of customers with substantial balances may represent greater potential financial exposure than the departure of customers with lower balances. Balance-based analysis can therefore complement customer-count churn analysis by identifying where the financial exposure associated with churn is concentrated. Salary can provide additional context about customer financial profiles, although a relationship between salary and balance should be empirically tested rather than assumed.

3.6 Analytical Gap and Relevance to This Study

A key practical requirement is to connect churn measurement with interpretable customer segments and financial exposure. An overall churn percentage can describe the scale of the problem, but it does not identify which segments warrant attention or whether high-value customers are disproportionately affected. This study addresses that requirement by integrating geographic, demographic, engagement, tenure, credit-score, and balance-based analysis within a single analytical framework. The findings are then translated into an interactive Streamlit dashboard so that users can explore customer segments and churn patterns dynamically.

Chapter 4. Research Methodology

This study uses a structured quantitative and descriptive analytical approach to examine customer churn patterns in a European banking dataset. The methodology is designed to directly address the project objectives of measuring overall churn, identifying churn distribution across customer segments, comparing churn across European regions, understanding high-value customer churn, evaluating engagement and tenure patterns, and supporting strategic planning and marketing decisions.

4.1 Research Approach

The study follows a customer-level observational analysis. Each row represents an individual customer, while the Exited variable represents the observed churn outcome. The analysis first validates and prepares the dataset, then creates interpretable customer segments, calculates churn rates and churn contributions, compares retained and churned customer profiles, and finally examines financial exposure associated with high-value churn.

The methodology is primarily descriptive and comparative rather than predictive. The purpose is to understand the distribution and characteristics of observed churn within the available customer population. Consequently, the findings are interpreted as associations in the dataset and are not presented as evidence of causal relationships.

4.2 Analytical Unit and Dataset Structure

The unit of analysis is the individual banking customer. The dataset contains 10,000 customer records and 14 original variables covering customer identification, demographics, financial characteristics, product ownership, engagement, and churn status. The primary outcome variable is Exited, where 1 represents a churned customer and 0 represents a retained customer.

Methodological element	Definition
Unit of analysis	Individual customer
Number of observations	10,000
Original variables	14
Outcome variable	Exited
Churn coding	1 = churned; 0 = retained
Geographic scope	France, Germany, Spain
Analytical approach	Descriptive and comparative analysis

4.3 Data Ingestion and Validation

The dataset was loaded into the Python analytical environment and systematically inspected before analysis. Validation focused on dataset dimensions, variable types, missing observations, duplicate records, categorical values, product fields, engagement fields, and the binary churn label.

The validation confirmed that NumOfProducts contains values from 1 to 4, while HasCrCard, IsActiveMember, and Exited contain only binary values of 0 and 1. No missing values or

duplicate rows were identified. The churn label was also checked for consistency before being used in subsequent calculations.

4.4 Data Cleaning and Preparation

Data preparation was undertaken to ensure that the variables used for analysis were relevant and consistently represented. The Surname variable was removed because it does not provide a meaningful analytical basis for customer segmentation or churn interpretation. CustomerId was retained for record identification but was not used as an analytical characteristic.

Because the dataset contained no missing observations, no imputation procedure was required. Existing binary variables were retained in their validated form. Categorical variables were prepared for grouping, while derived segmentation variables were created from age, credit score, tenure, and balance.

4.5 Customer Segmentation Method

Customer segmentation was designed to provide interpretable categories for comparing churn behaviour. Five dimensions were used: geography, age, credit score, tenure, and balance. The thresholds were defined before the comparative churn analysis and were applied consistently across the dataset.

Segmentation dimension	Categories	Operational definition
Geography	France; Germany; Spain	Existing geography classification
Age	Under 30; 30–45; 46–60; 60+	Age bands
Credit Score	Low; Medium; High	Low <580; Medium 580–669; High ≥ 670
Tenure	New; Mid-term; Long-term	New 0–3; Mid-term 4–7; Long-term 8–10
Balance	Zero Balance; Low Balance; High Balance	0; >0 and <100,000; $\geq 100,000$

4.6 Churn Measurement

Churn was measured using the Exited variable. The overall churn rate was calculated as the number of churned customers divided by the total number of customers, multiplied by 100. Segment-level churn rates were calculated using the number of churned customers within each segment divided by the total number of customers in that segment, multiplied by 100.

In addition to churn rates, churn contribution was calculated to determine the proportion of all churned customers belonging to each segment. This distinction allows the study to separate the risk level within a segment from the absolute number of churners contributed by that segment.

Measure	Calculation concept	Purpose
Overall churn rate	Total churned / total customers $\times$ 100	Measure overall churn level
Segment churn rate	Segment churned / segment customers $\times$ 100	Compare relative churn risk across groups
Churn contribution	Segment churners / total churners $\times$ 100	Measure each segment's share of all churners

4.7 Comparative Customer Profile Analysis

Retained and churned customers were compared using numerical and categorical characteristics. For numerical variables, mean and median values were examined for CreditScore, Age, Tenure, Balance, NumOfProducts, and EstimatedSalary. For categorical variables, the composition of retained and churned populations was compared across geography, gender, credit-card ownership, and active-member status.

This comparison provides context for the segment-level churn rates and helps identify characteristics that are disproportionately represented among churned customers.

4.8 Demographic and Geographic Interaction Analysis

A combined geography-by-age analysis was performed to identify whether churn patterns differed when geographic and demographic characteristics were considered together. Churn rates were calculated for each geography and age-segment combination. This approach helps identify high-risk combinations that may not be visible from separate geography-only and age-only analyses.

4.9 High-Value Customer Analysis

High-value customers were operationally represented by the High Balance segment, defined as customers with a balance of 100,000 or above in the dataset's balance units. The analysis identified the number of high-balance churners and examined their balance characteristics using total, average, and median balance.

Salary and balance were also examined among high-balance churners using a scatter plot and Pearson correlation coefficient. The correlation coefficient was used to assess the strength of the linear relationship between the two numerical variables. A correlation value close to zero was interpreted as evidence of little linear association, not as evidence that the variables are completely unrelated.

4.10 Financial Exposure Analysis

Financial exposure associated with churn was assessed by summing the account balances of churned customers. This provides a balance-based measure of the amount of customer balance associated with observed churn. The analysis further separates high-balance churners from other churned customers to quantify the concentration of this exposure.

This measure is deliberately described as balance exposure rather than lost revenue. The dataset does not contain sufficient information about fees, interest income, profitability, customer lifetime value, or other revenue streams to calculate actual revenue loss. Therefore, the balance-based measure is treated as a proxy for potential financial risk associated with customer churn.

4.11 Analytical Tools and Implementation

The analysis was implemented in Python using a notebook-based workflow. Data preparation and analysis were conducted using pandas and NumPy, while matplotlib was used to produce analytical visualisations. The final findings were organised into tables and charts for interpretation and were subsequently incorporated into an interactive Streamlit dashboard.

Tool / library	Methodological role
Python	Primary analytical programming environment
pandas	Data loading, cleaning, transformation and grouping
NumPy	Numerical operations and analytical support
Matplotlib	Charts and analytical visualisations
Streamlit	Interactive presentation of analytical findings

4.12 Interpretation and Ethical Considerations

The analysis is based on observed customer characteristics and churn outcomes. Segment differences are therefore interpreted as statistical associations within the dataset rather than causal effects. In particular, demographic variables such as age, gender, and geography should not be interpreted as direct causes of churn. Similarly, financial exposure associated with churn should not be interpreted as realised financial loss.

The methodology prioritises aggregated segment-level interpretation rather than exposing individual customer information. Customer names and surnames are not used in the analysis, and CustomerId is treated only as an identifier. This supports a more appropriate analytical focus on patterns and segments rather than individual-level identification.

4.13 Methodological Workflow

1. Load and inspect the European banking dataset.
2. Validate data structure, missing values, duplicates, product fields, engagement fields, and churn labels.
3. Remove non-analytical fields and prepare variables for analysis.
4. Create geographic, age, credit-score, tenure, and balance segments.
5. Calculate overall and segment-wise churn rates.
6. Calculate churn contribution by segment size.
7. Compare retained and churned customer profiles.
8. Examine gender and geography-age churn patterns.
9. Identify high-balance churners and quantify balance exposure.
10. Assess the salary-balance relationship among high-balance churners.
11. Translate findings into visualisations and an interactive Streamlit dashboard.

The methodology therefore provides a consistent analytical pathway from raw customer records to interpretable churn findings, while maintaining a clear distinction between descriptive evidence, financial exposure, and causal or revenue-based claims.

Chapter 5. Data Description and Preparation

5.1 Dataset Description

The study uses a customer-level European banking dataset containing 10,000 observations and 14 original variables. The dataset captures demographic, financial, product, engagement and churn information. The target outcome is Exited, where 1 denotes a churned customer and 0 denotes a retained customer.

Initial validation found no missing values and no duplicate records. CustomerId is unique across the 10,000 observations. The Year field contains one observed year and therefore does not provide a meaningful basis for time-series comparison in this study.

5.2 Data Quality and Cleaning

Data preparation focused on ensuring that analytical variables were complete, consistent and relevant. The Surname field was removed because it has no meaningful analytical role in customer segmentation. CustomerId was retained only as an identifier. Binary fields were checked for invalid values, and no inconsistencies were found.

Check	Result
Records	10,000
Original variables	14
Missing values	None
Duplicate rows	None
Invalid binary values	None
Cleaned variables	13

5.3 Customer Population Structure

The customer population is geographically concentrated in France, which accounts for 50.14% of customers. The 30–45 age group is the largest age category at 62.48%, while High credit-score customers represent 43.07% of the population. High Balance customers account for 47.99% of customers.

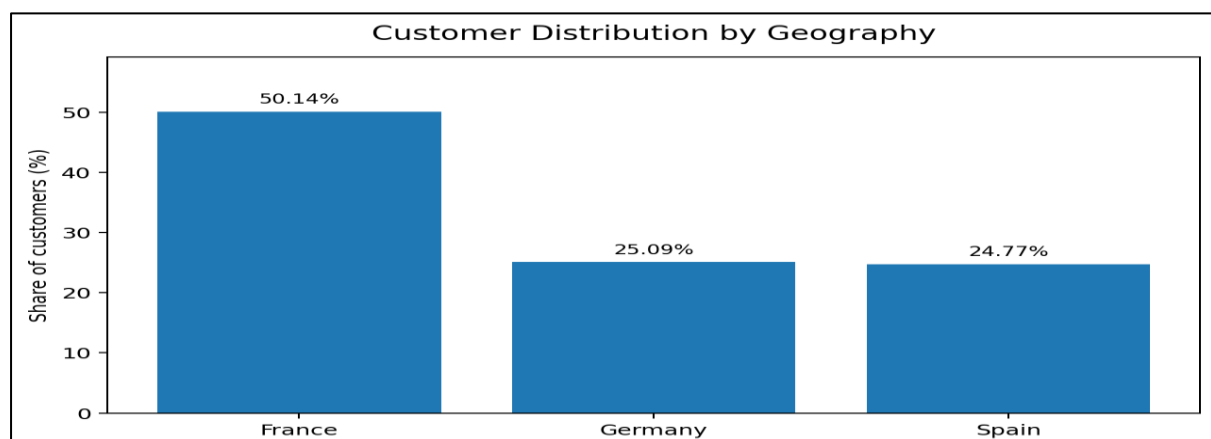


Figure 5.1. Customer distribution by geography.

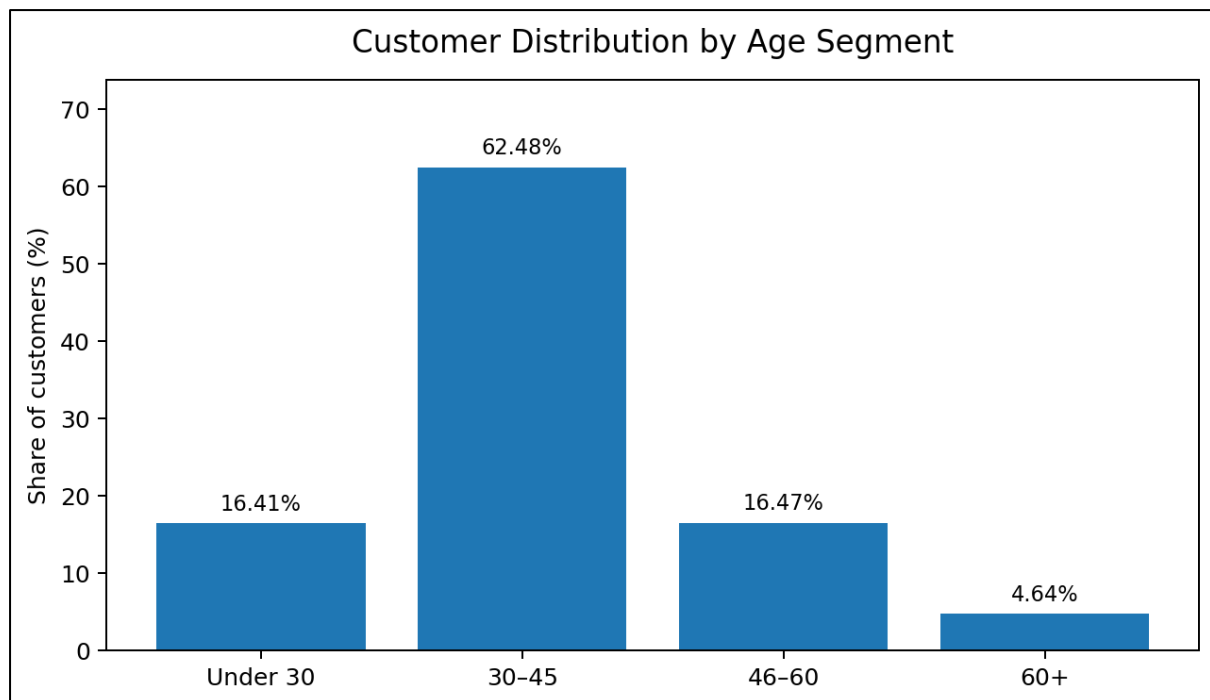


Figure 5.2. Customer distribution by age segment.

These distributions are important when interpreting subsequent churn results because segment size affects the number of churners contributed by each group.

5.4 Numerical Customer Profile

Credit scores range from 350 to 850, age ranges from 18 to 92, and tenure ranges from 0 to 10 years. Balance ranges from zero to approximately 250,898, while EstimatedSalary ranges from approximately 11.58 to 199,925. These ranges provide the basis for the segmentation thresholds described in the next section.

Variable	Minimum	Median	Maximum
CreditScore	350	652	850
Age	18	37	92
Tenure	0	5	10
Balance	0	≈97,185	≈250,898
NumOfProducts	1	1	4
EstimatedSalary	≈11.58	≈100,194	≈199,925

Chapter 6. Customer Segmentation

Five segmentation dimensions were applied consistently: geography, age, credit score, tenure and balance. The categories were selected to provide interpretable groups for direct comparison of customer populations and churn behaviour.

6.1 Segmentation Framework

Dimension	Categories	Operational rule
Geography	France; Germany; Spain	Existing categorical field
Age	Under 30; 30–45; 46–60; 60+	Predefined age bands
Credit Score	Low; Medium; High	<580; 580–669; ≥ 670
Tenure	New; Mid-term; Long-term	0–3; 4–7; 8–10
Balance	Zero; Low; High	0; >0 and <100,000; $\geq 100,000$

6.2 Credit Score Segmentation

The High credit-score group is the largest segment, containing 4,307 customers. Medium and Low groups contain 3,331 and 2,362 customers respectively.

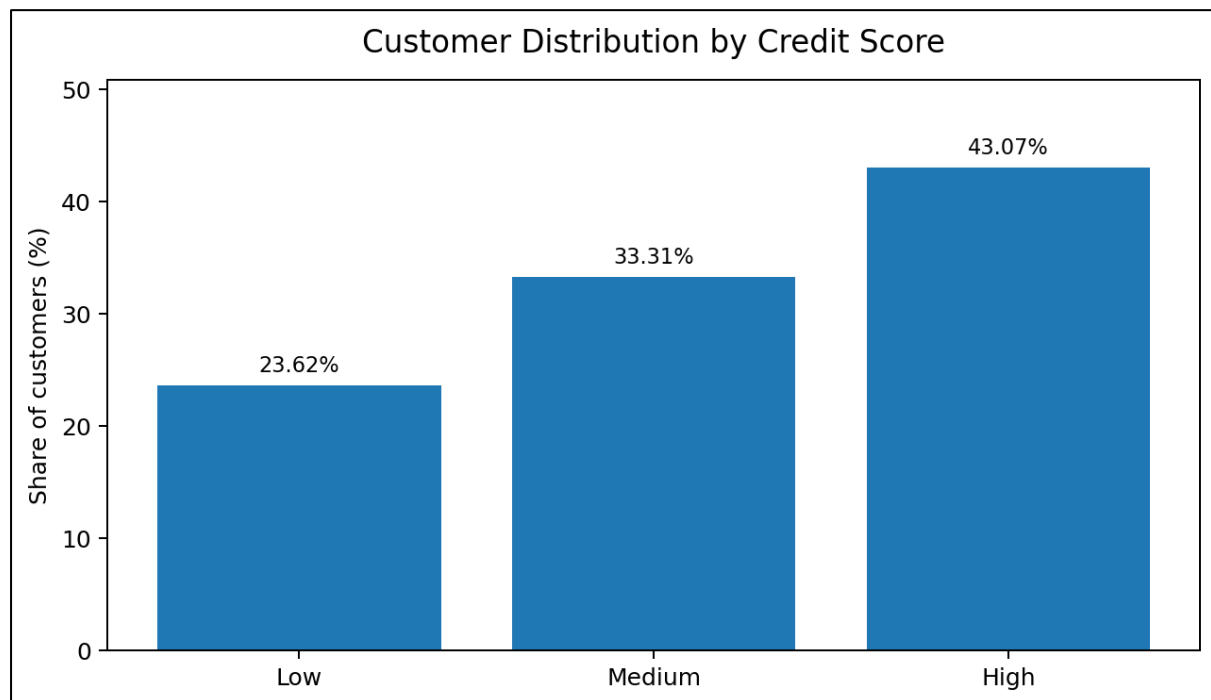


Figure 6.1. Customer distribution by credit-score segment.

6.3 Tenure Segmentation

Mid-term customers form the largest tenure group with 3,996 customers, followed by new customers with 3,505 and Long-term customers with 2,499.

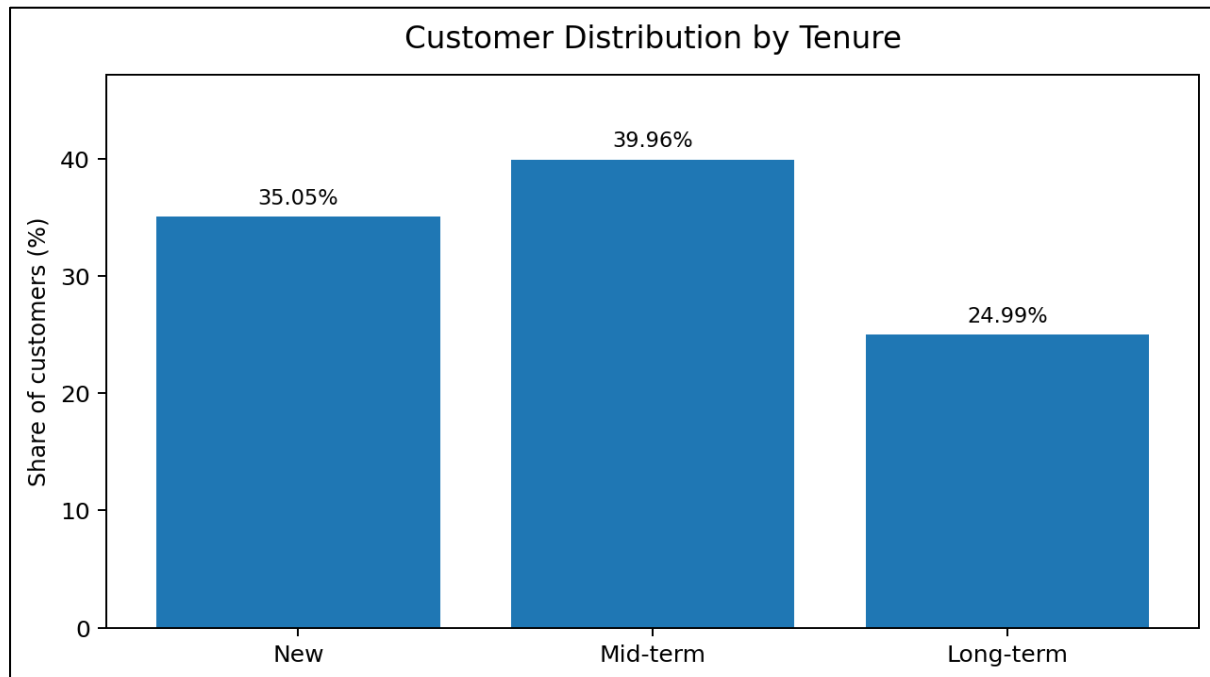


Figure 6.2. Customer distribution by tenure segment.

6.4 Balance Segmentation

High Balance is the largest balance segment with 4,799 customers, followed by Zero Balance with 3,617 and Low Balance with 1,584. This segmentation is particularly important because balance is later used to identify high-value churners and quantify balance exposure associated with churn.

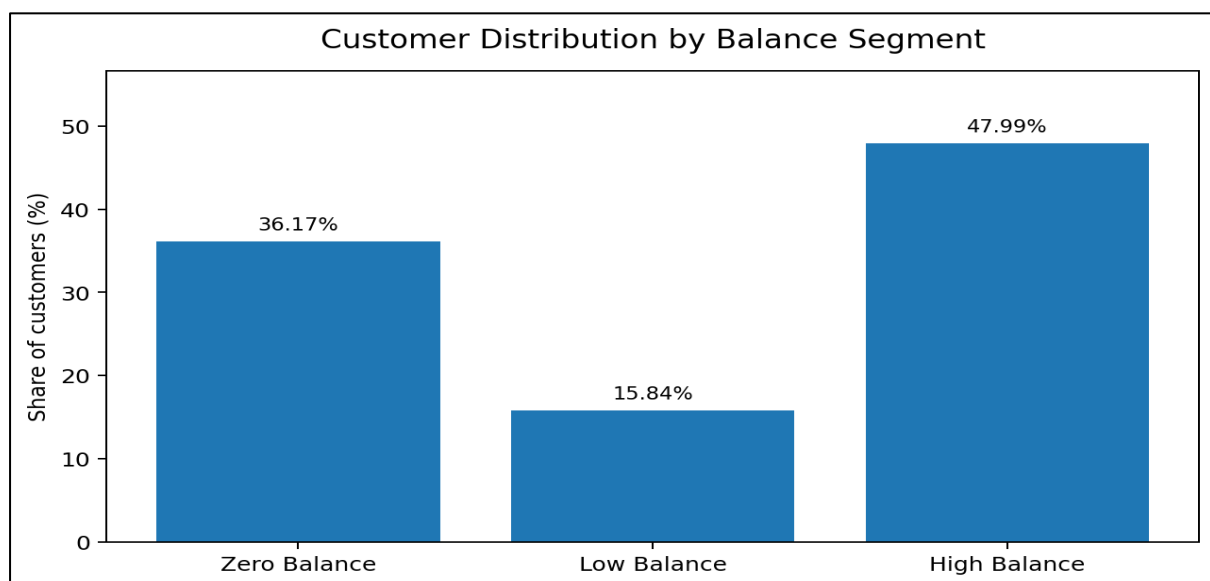


Figure 6.3. Customer distribution by balance segment.

Chapter 7. Churn Analysis and Results

The churn analysis evaluates the scale and distribution of observed churn across the customer population. Churn rates are interpreted alongside segment size so that high-risk groups are not confused with groups that simply contain many customers.

7.1 Overall Churn

Of the 10,000 customers, 2,037 churned and 7,963 were retained, resulting in an overall churn rate of 20.37%.

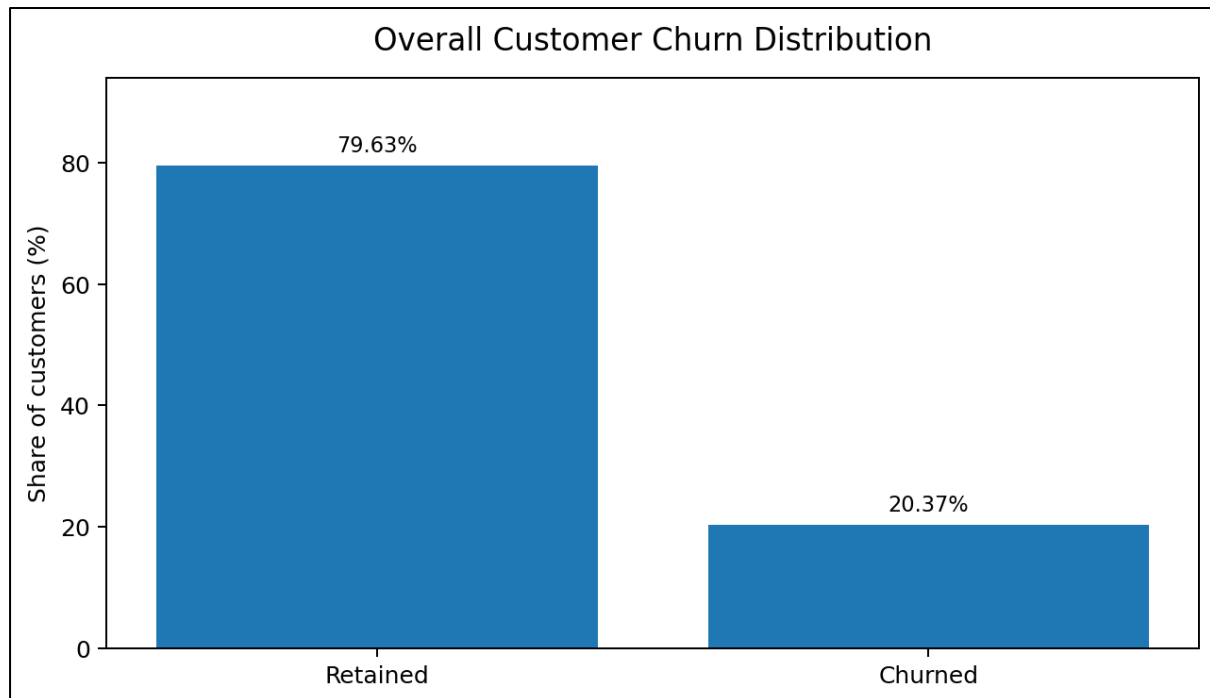


Figure 7.1. Overall customer churn distribution.

Status	Customers	Share
Retained	7,963	79.63%
Churned	2,037	20.37%

7.2 Geographic Churn

Germany has the highest geographic churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain. Germany therefore stands out as the highest-risk geographic segment in the observed dataset.

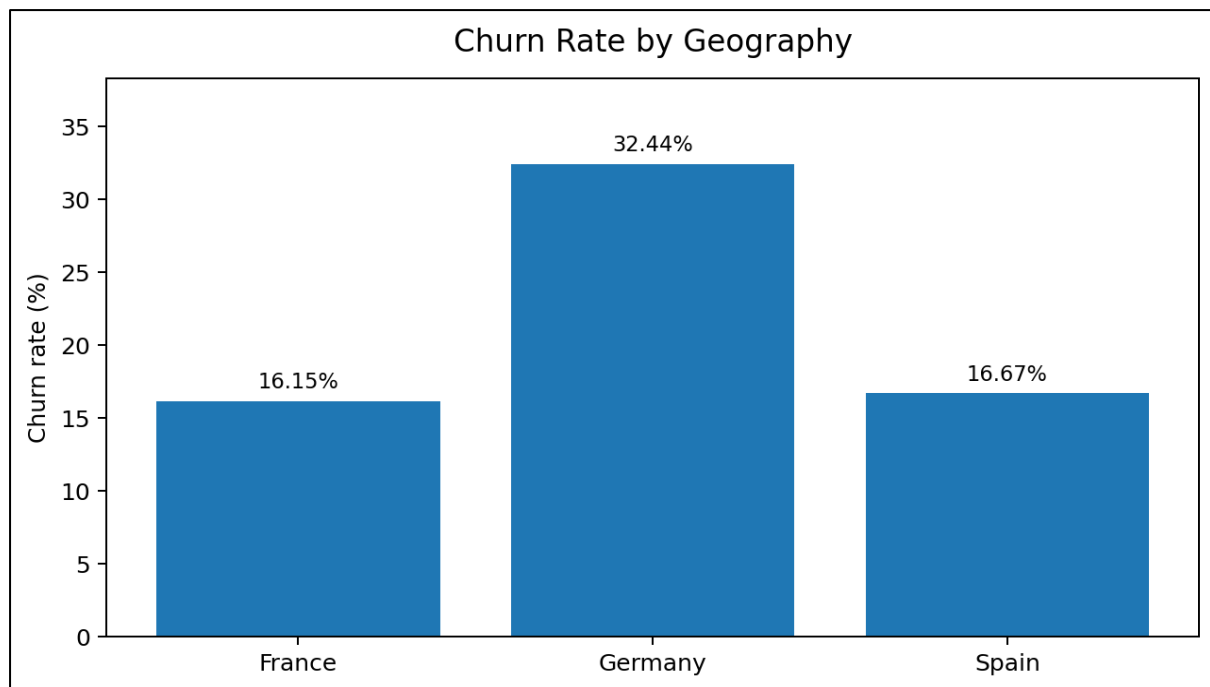


Figure 7.2. Churn rate by geography.

Although France has the largest customer population, Germany contributes 814 churners, representing 39.96% of all churners. This illustrates why both churn rate and churn contribution should be considered.

7.3 Age-Based Churn

Age produces one of the strongest observed differences in churn. The 46–60 group records a churn rate of 51.12%, followed by 60+ at 24.78%. The 30–45 group records 15.30%, while Under 30 has the lowest rate at 7.56%.

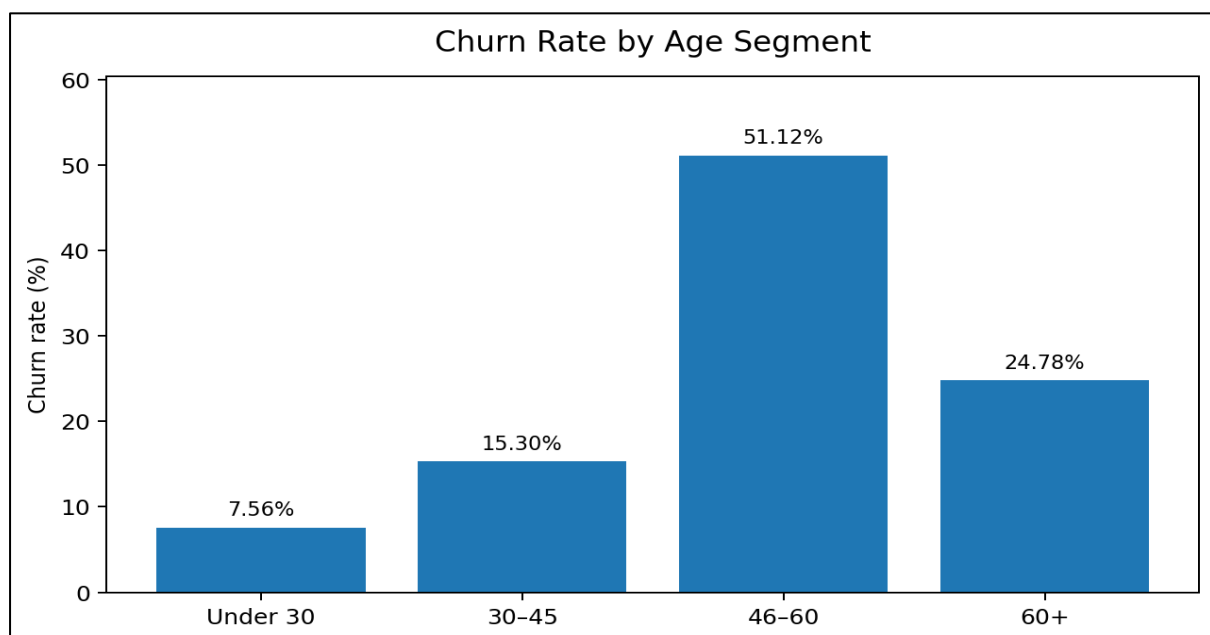


Figure 7.3. Churn rate by age segment.

7.4 Gender-Based Churn

Female customers show a churn rate of 25.07%, compared with 16.46% among male customers. The difference is 8.61 percentage points. This is an observed demographic difference and should not be interpreted as evidence that gender causes churn.

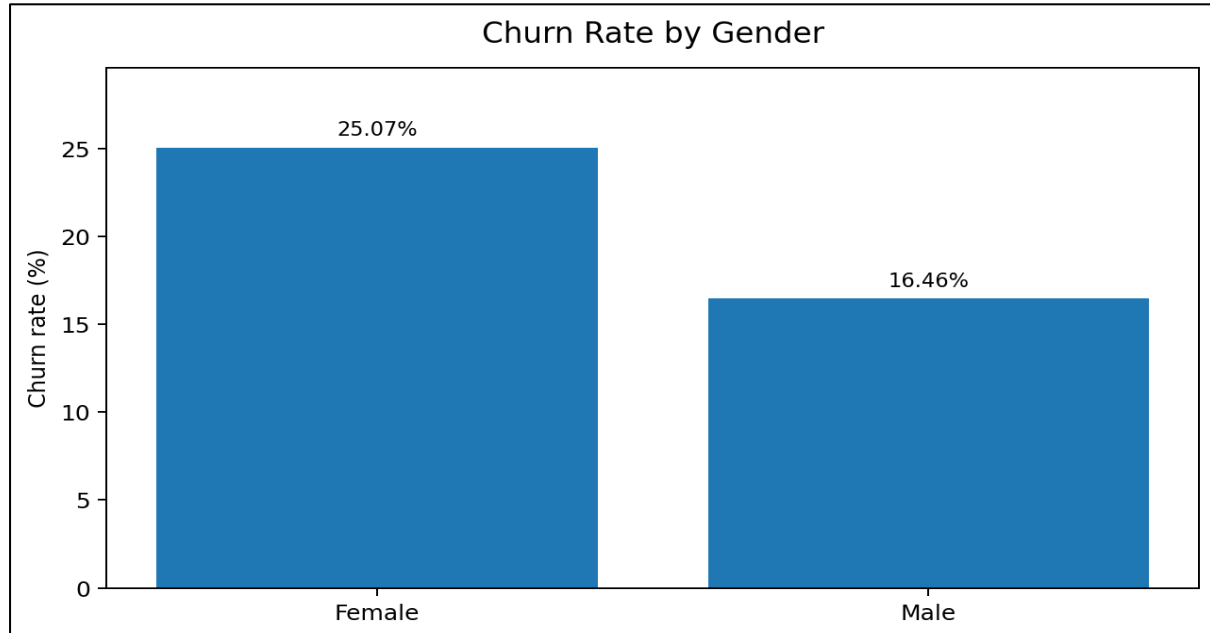


Figure 7.4. Churn rate by gender.

7.5 Balance-Based Churn

High-balance customers have the highest balance-segment churn rate at 25.23%, compared with 20.58% for Low Balance and 13.82% for Zero Balance. High-balance customers also account for 59.45% of all churners.

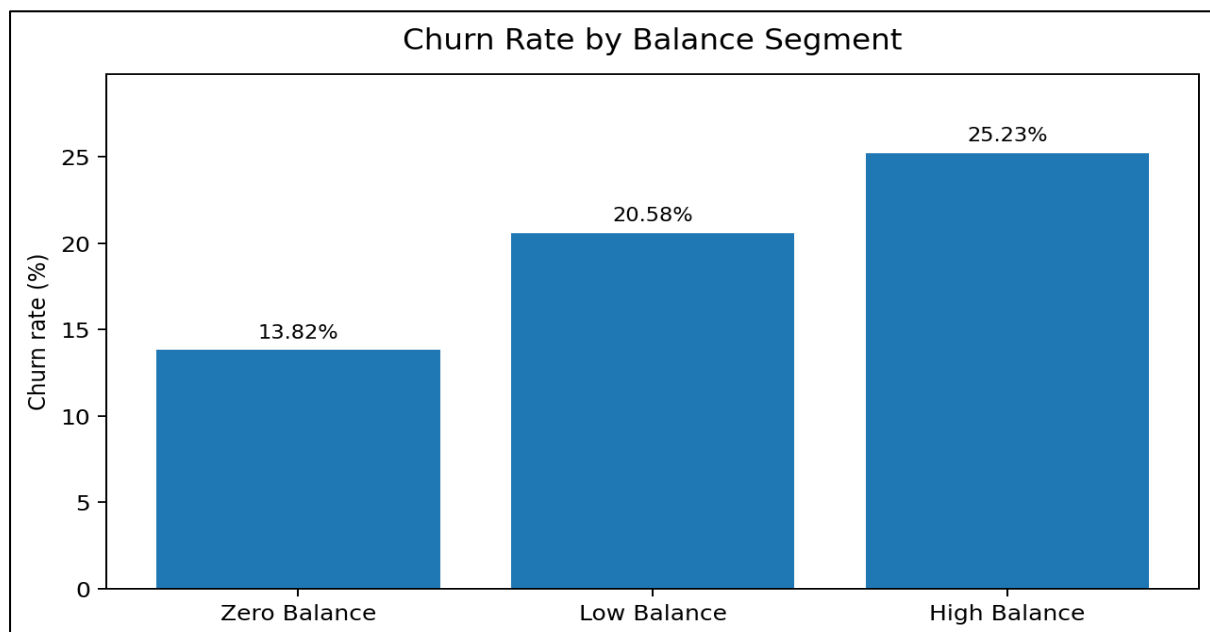


Figure 7.5. Churn rate by balance segment.

7.6 Geography–Age Interaction

The interaction between geography and age reveals a particularly high-risk combination. Germany customers aged 46–60 have the highest observed churn rate at 67.33%. The corresponding rate is 45.79% in France and 40.66% in Spain. Germany also records elevated churn across the other age groups.

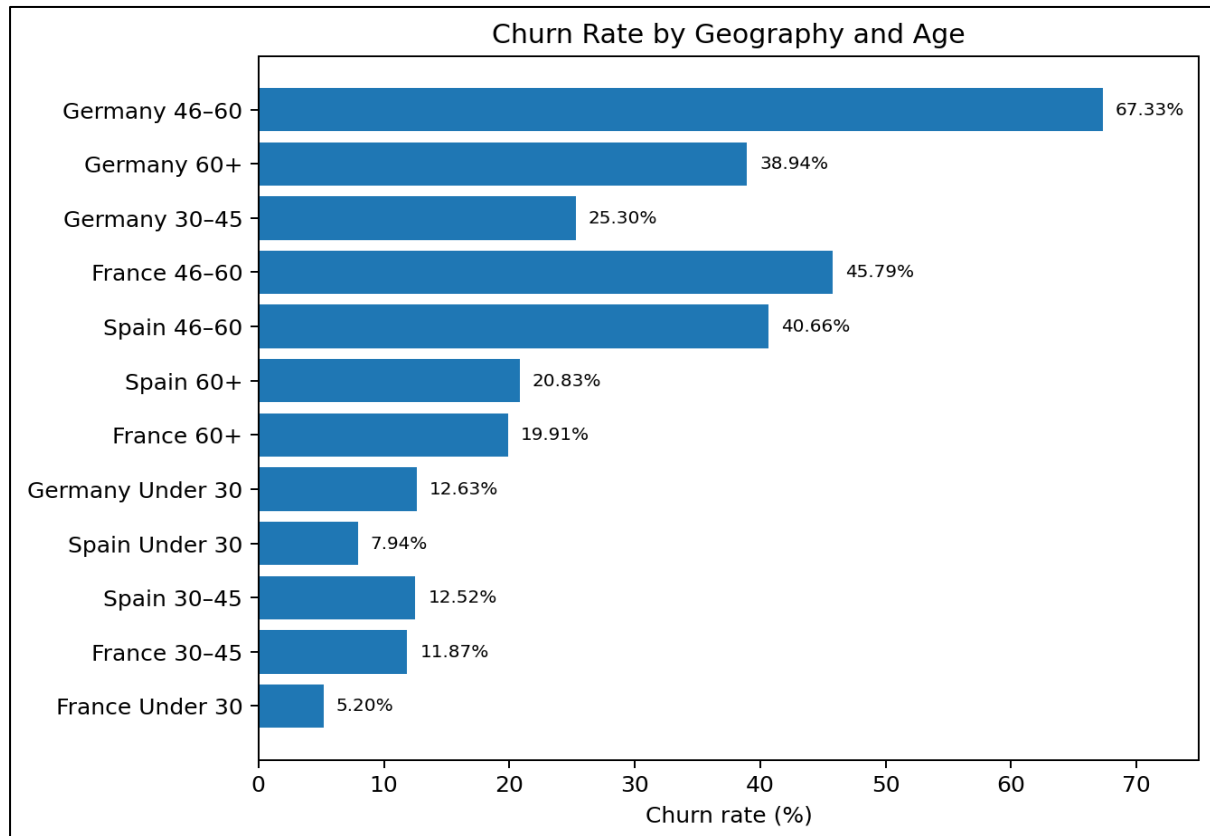


Figure 7.6. Churn rate by geography and age. The Germany 46–60 segment records the highest observed rate at 67.33%.

7.7 Retained and Churned Customer Profiles

Churned customers are older and have higher balances on average than retained customers. Average age is 44.84 for churned customers compared with 37.41 for retained customers. Average balance is 91,108.54 among churners compared with 72,745.30 among retained customers. Average credit score is slightly lower among churners, while average tenure is almost identical.

Metric	Retained mean	Churned mean
Credit Score	651.85	645.35
Age	37.41	44.84
Tenure	5.03	4.93
Balance	72,745.30	91,108.54
NumOfProducts	1.54	1.48
Estimated Salary	99,738.39	101,465.68

7.8 Engagement and Product Characteristics

Inactive customers account for 63.92% of churners, compared with 44.54% of retained customers. Active customers account for 36.08% of churners compared with 55.46% of retained customers. Credit-card ownership is comparatively similar between groups, with 69.91% of churners and 70.71% of retained customers holding a credit card.

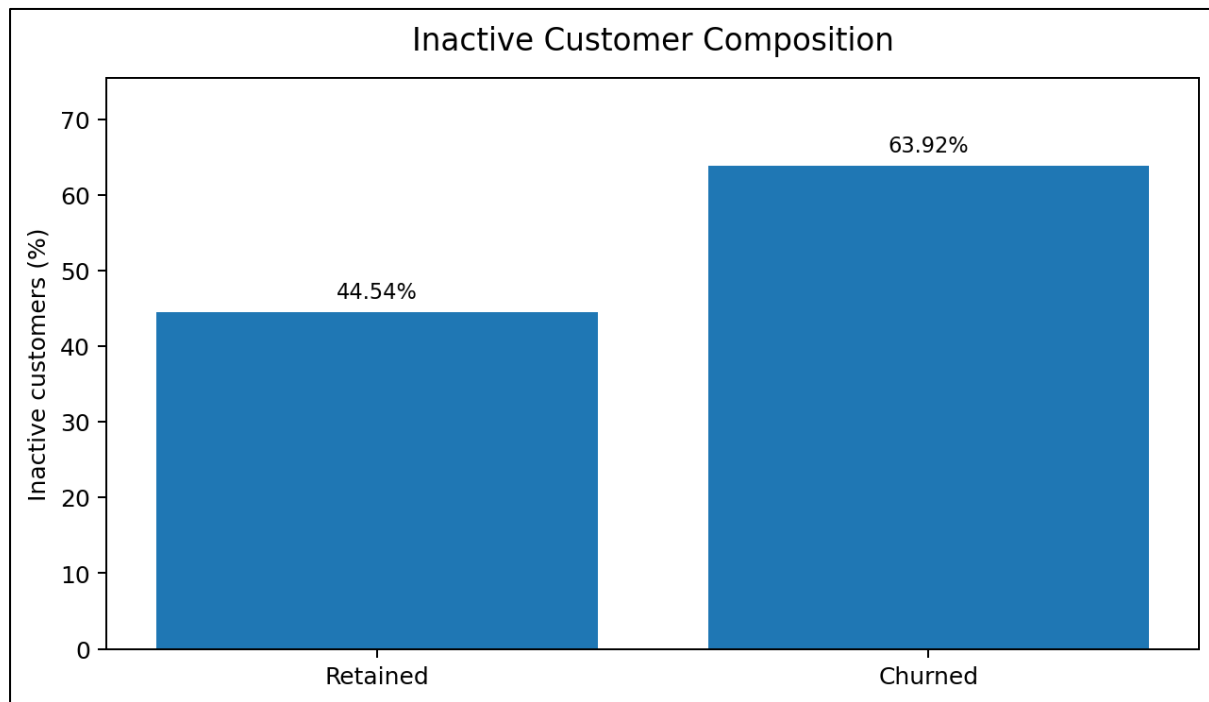


Figure 7.7. Inactive customer composition among retained and churned populations.

The engagement result provides a stronger observed distinction than credit-card ownership. It suggests that activity status is an important descriptive characteristic for further retention analysis.

7.9 Credit Score and Tenure Findings

Credit-score churn rates range from 19.32% for High credit score to 22.02% for Low credit score. Tenure churn rates range from 19.64% for Mid-term customers to 21.14% for New customers. The relatively narrow differences indicate that these variables are weaker differentiators of churn in this dataset than geography, age, balance and engagement.

7.10 Summary of Results

The results show that churn is concentrated in specific customer segments rather than being evenly distributed. Germany, customers aged 46–60, female customers, high-balance customers, and inactive customers display notable differences in observed churn patterns. The strongest combined segment is Germany customers aged 46–60, with a churn rate of 67.33%. These findings form the basis for the high-value customer and financial-risk analysis in Section-8.

Chapter 8. High-Value Customer Churn and Financial Risk

High-value customer churn is examined separately because customer count alone does not capture the financial significance of churn. In this study, high-value customers are operationally represented by the High Balance segment, defined as customers with a balance of 100,000 or above in the dataset's balance units.

8.1 Identification of High-Balance Churners

A total of 1,211 high-balance customers were identified among the churned population. These customers represent 59.45% of all churners. Their average balance is 131,700.82 and their median balance is 127,209.00.

Measure	Result
High-balance churners	1,211
Share of all churners	59.45%
Total balance exposure	159,489,691.00
Average balance	131,700.82
Median balance	127,209.00

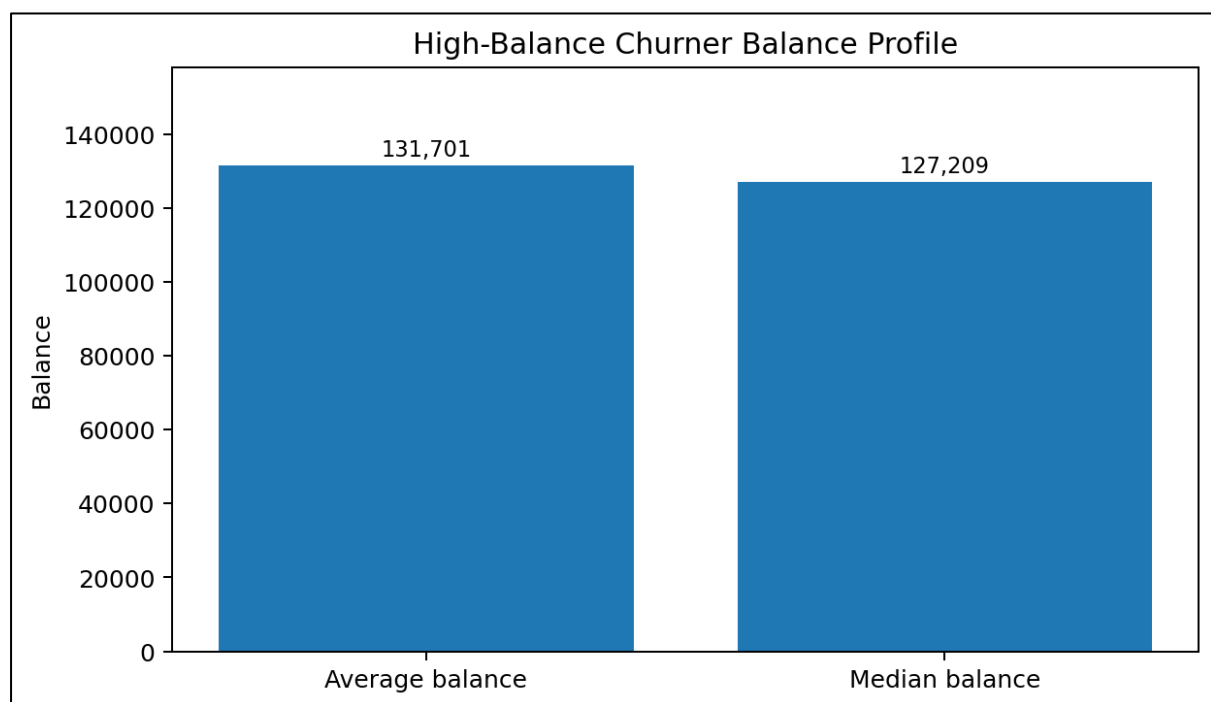


Figure 8.1. Average and median balance among high-balance churners.

8.2 Salary and Balance Relationship

The relationship between estimated salary and balance was examined among high-balance churners using a scatter plot and Pearson correlation. The resulting correlation coefficient is 0.0123, indicating an essentially negligible linear relationship between the two variables within this group.

This result suggests that higher estimated salary is not linearly associated with higher balance among the high-balance customers who churned in this dataset. However, correlation does not establish causation and a near-zero Pearson coefficient does not rule out non-linear or more complex relationships.

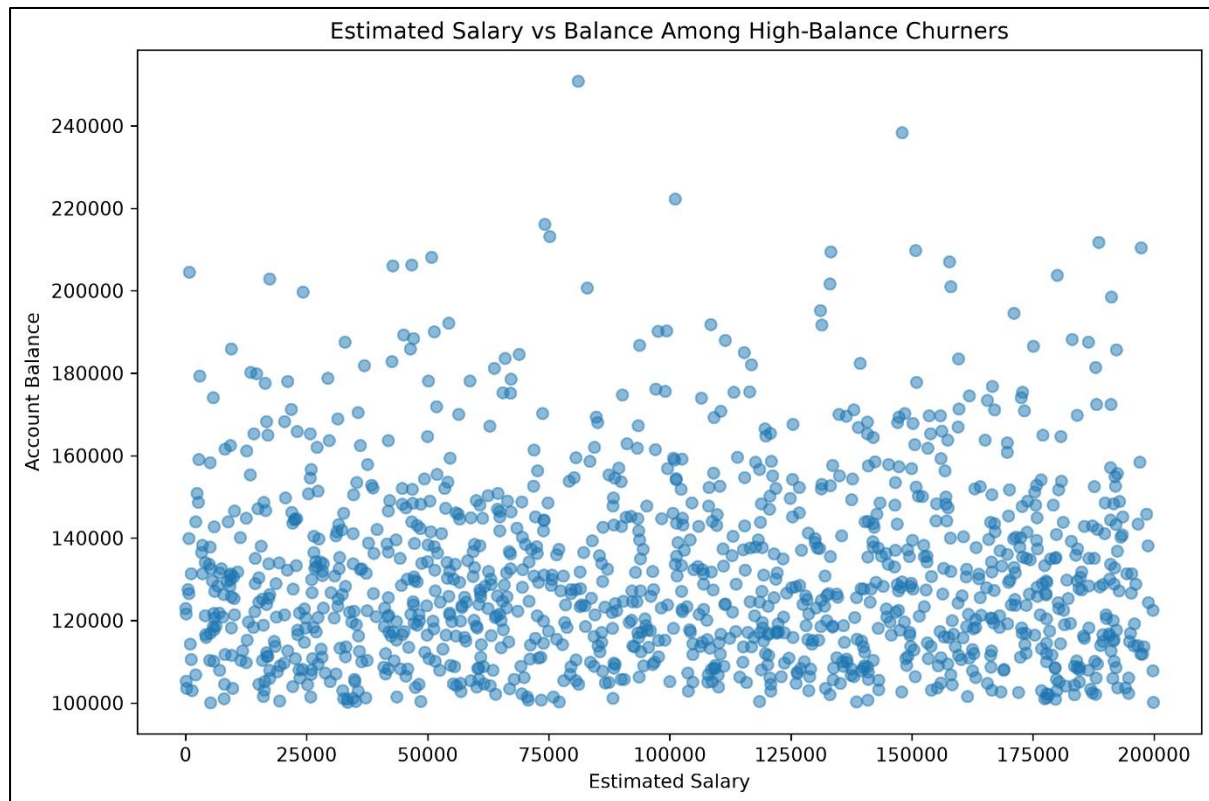


Figure 8.2. Salary vs Balance among high-balance churners. The project analysis reports Pearson $r = 0.0123$, indicating negligible linear association.

8.3 Implications of High-Value Churn

The concentration of balance exposure among high-balance churners indicates that retention analysis should consider both the probability of churn and the financial profile of the customers affected. A strategy based only on the number of customers at risk could overlook the substantially greater balance exposure associated with some churned customers. Accordingly, high-balance segments represent an important population for targeted retention investigation.

8.4 Balance Exposure Associated with Churn

The total balance associated with all churned customers is 185,588,094.63. Of this amount, 159,489,691.00 is associated with high-balance churners, while 26,098,403.63 is associated with other churned customers. High-balance churners therefore account for 85.94% of the total balance exposure associated with churn.

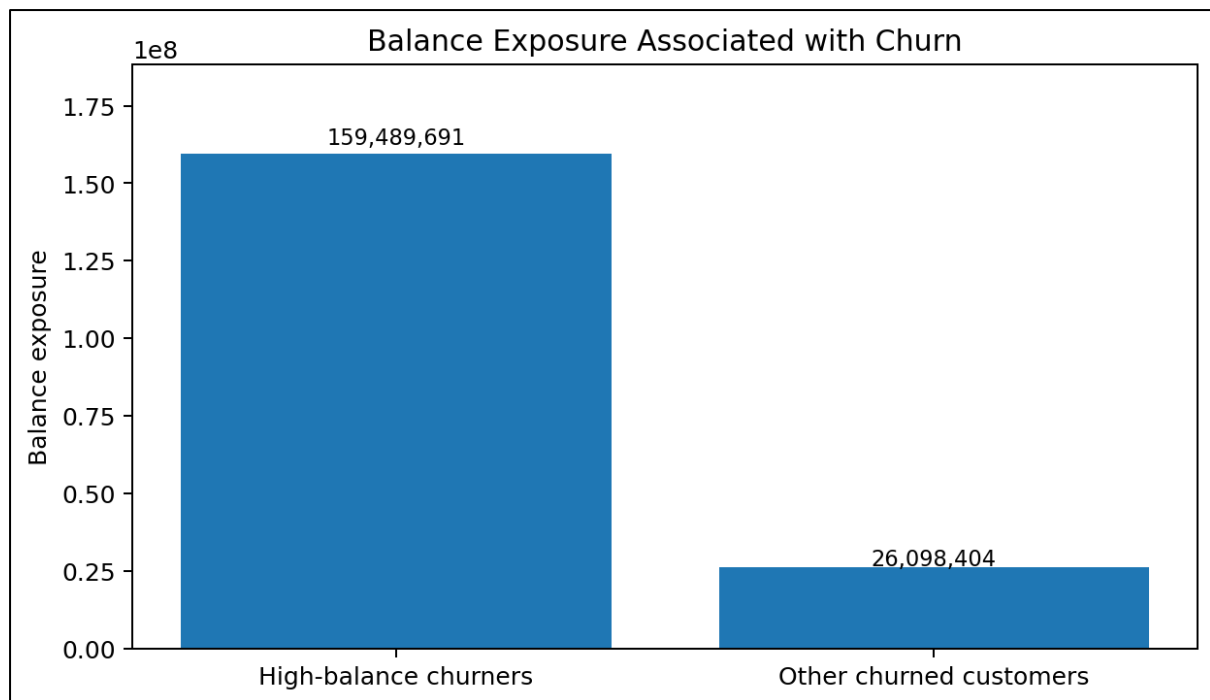


Figure 8.3. Balance exposure associated with churn. High-balance churners account for the majority of the observed exposure.

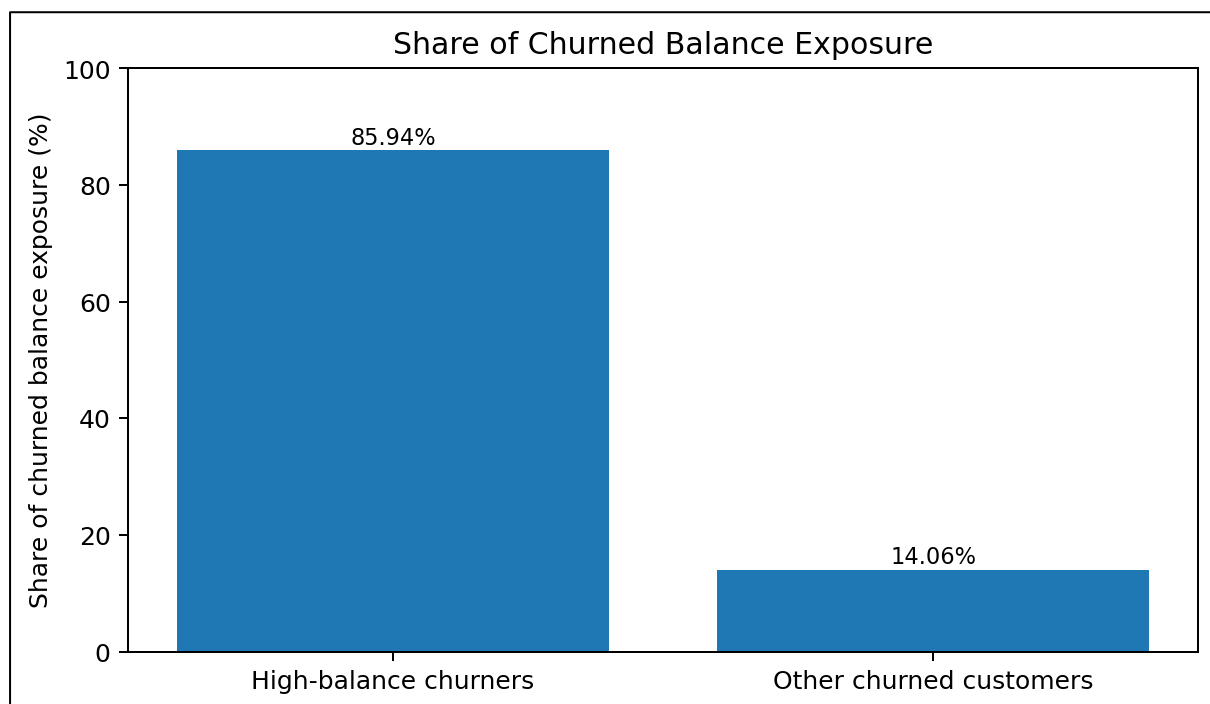


Figure 8.4. Share of total balance exposure associated with churn from high-balance and other churned customers.

The exposure measure should not be interpreted as actual lost revenue. The dataset does not contain information required to estimate lost interest income, fees, profitability, or customer lifetime value. Instead, balance exposure provides a proxy for the scale of financial resources associated with customers who have churned.

Chapter 9. Interactive Streamlit Dashboard

To translate the analytical findings into a practical decision-support format, an interactive Streamlit dashboard was developed. The dashboard allows users to explore churn patterns dynamically by applying customer-segment filters and viewing key performance indicators, charts, and high-value customer risk measures.

9.1 Dashboard Purpose

The dashboard is designed to make the analytical results accessible without requiring users to work directly with the underlying dataset or Python notebook. It provides a single interface for examining overall churn, segment-level risk, customer composition, engagement differences, and financial exposure associated with churn.

9.2 Interactive Filtering

Users can filter the customer population using Geography, Age Segment, Credit Score, Tenure, and Balance Segment. The displayed metrics and analytical views respond to the selected filters, allowing users to examine specific customer groups rather than relying only on aggregate results.

Dashboard filter	Available categories
Geography	All; France; Germany; Spain
Age Segment	All; Under 30; 30–45; 46–60; 60+
Credit Score	All; Low; Medium; High
Tenure	All; New; Mid-term; Long-term
Balance	All; Zero Balance; Low Balance; High Balance

9.3 Key Performance Indicators

The dashboard includes four principal analytical indicators. Overall Churn Rate measures churn across the selected population. Segment Churn Rate measures churn within the currently filtered population. High-Value Churn Ratio measures high-balance churners relative to high-balance customers. Geographic Risk Index compares the selected geography's churn rate with the overall churn rate, with the overall population represented as an index of 100.

Indicator	Definition
Overall Churn Rate	Churned customers ÷ all customers
Segment Churn Rate	Churned customers ÷ filtered customers
High-Value Churn Ratio	High-balance churners ÷ high-balance customers
Geographic Risk Index	Selected geography churn rate ÷ overall churn rate × 100

Chapter 10. Discussion

The findings demonstrate that customer churn is unevenly distributed across the banking customer population. The overall churn rate of 20.37% provides a useful baseline, but the segment-level analysis reveals substantially higher observed churn in particular demographic, geographic, engagement, and financial groups.

10.1 Geographic and Demographic Patterns

Germany records the highest geographic churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain. More importantly, the geography-by-age interaction identifies Germany customers aged 46–60 as the highest observed risk combination, with a churn rate of 67.33%. This demonstrates the value of combining segmentation dimensions rather than examining each characteristic independently.

10.2 Age and Gender Differences

Age is one of the strongest differentiators in the analysis. The 46–60 segment has a churn rate of 51.12%, substantially above the overall rate of 20.37%. Female customers also show a higher churn rate than male customers, 25.07% compared with 16.46%. These differences can support targeted investigation, but they should not be interpreted as causal effects of age or gender.

10.3 Engagement and Tenure

Engagement status shows a notable difference between retained and churned populations. Inactive customers account for 63.92% of churners compared with 44.54% of retained customers. This suggests that activity status is a useful descriptive indicator of customer disengagement. By contrast, tenure shows relatively limited variation, with churn rates ranging from 19.64% among Mid-term customers to 21.14% among New customers.

10.4 Financial Characteristics and High-Value Churn

High-balance customers have a churn rate of 25.23% and account for 59.45% of all churners. More importantly, high-balance churners account for 85.94% of the balance exposure associated with all churned customers. This distinction between customer count and financial exposure is important because the business significance of churn may differ according to the financial profile of affected customers.

10.5 Relative Strength of Findings

Taken together, geography, age, balance, gender, and engagement show stronger observed differences than credit score, tenure, or credit-card ownership. Credit-card ownership is almost identical between retained and churned customers, while credit-score churn rates differ only modestly. These results suggest that a broad retention strategy based on a single customer characteristic would be less informative than a segmented approach that considers multiple dimensions.

10.6 From Descriptive Evidence to Business Action

The findings provide a basis for prioritising further retention investigation. The highest-risk combinations identified in the data can be examined more closely, while high-balance churners can be prioritised when the objective is to protect customer balance exposure. The Streamlit dashboard extends this analysis by enabling users to explore these patterns interactively.

10.7 Interpretation Boundaries

The discussion should be interpreted within the limits of the dataset and methodology. The analysis identifies associations in observed customer data rather than causal relationships. Balance exposure is a proxy for potential financial risk and is not equivalent to actual lost revenue. Similarly, high churn in a segment does not by itself establish why customers left or which intervention would successfully retain them.

Chapter 11. Strategic Recommendations

The analytical findings provide a basis for targeted retention planning. The recommendations below are derived from observed patterns in the dataset and are intended to support prioritisation rather than prescribe causal interventions. Because the analysis is descriptive, each recommendation should be validated through operational testing before implementation.

11.1 Prioritise High-Risk Geographic and Age Segments

Germany should receive particular analytical attention because it has the highest geographic churn rate at 32.44%. Within the geography-age interaction, customers aged 46–60 in Germany show the highest observed churn rate at 67.33%. A practical retention strategy could therefore begin with targeted investigation of this combined segment, including review of engagement, product usage, service interactions, and customer feedback.

11.2 Develop Targeted Retention for High-Balance Customers

High-balance customers have a churn rate of 25.23% and account for 59.45% of all churners. More importantly, high-balance churners account for 85.94% of the total balance exposure associated with churn. This suggests that retention planning should incorporate customer financial value alongside customer count when determining intervention priorities.

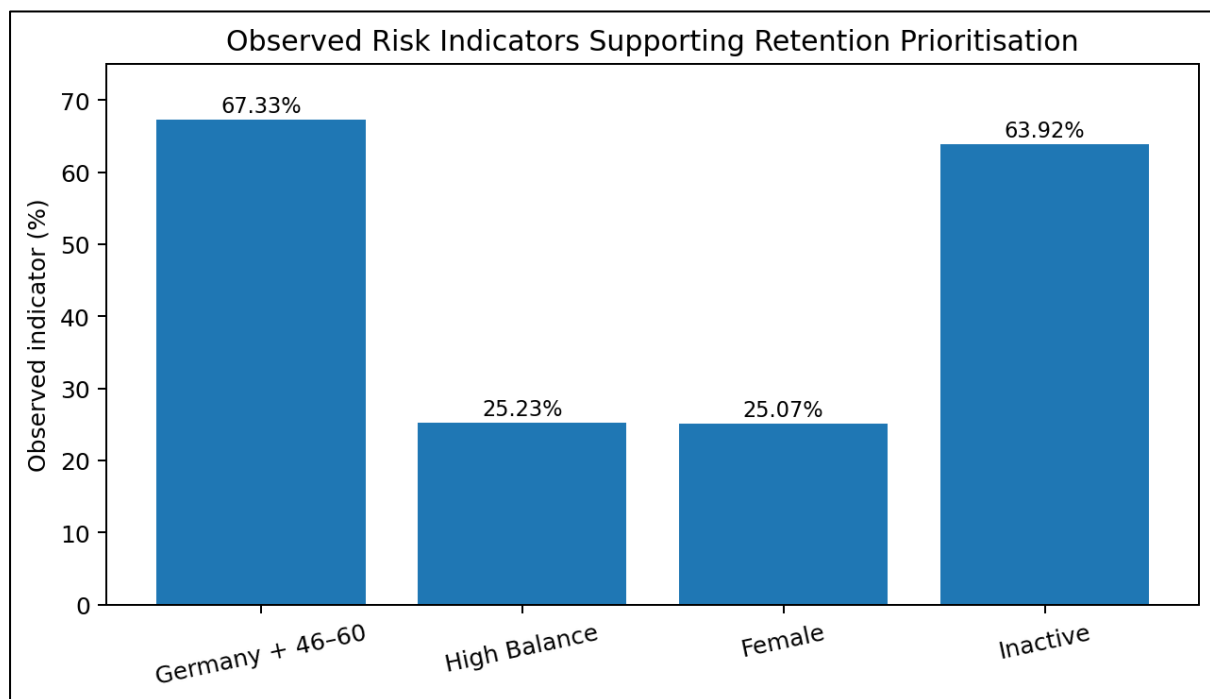


Figure 11.1. Observed indicators supporting retention prioritisation. The values represent different analytical measures and should not be interpreted as a common risk score.

11.3 Strengthen Engagement Monitoring

Inactive customers represent 63.92% of churners compared with 44.54% of retained customers. This difference supports the use of engagement status as an early descriptive indicator for

retention monitoring. Customers showing reduced activity could be considered for proactive engagement, subject to appropriate business rules and customer consent.

11.4 Use Segmentation Rather Than Generic Retention

The variation in churn across geography, age, gender, balance, and engagement demonstrates that a single retention approach may not address the needs of all customers equally. Retention activity should therefore be segmented according to observed customer characteristics. For example, high-balance customers may require value-focused retention analysis, while inactive customers may require engagement-focused interventions.

11.5 Avoid Over-Prioritising Weak Differentiators

Credit score and tenure show relatively limited differences in churn rates, while credit-card ownership is broadly similar between retained and churned customers. These variables should therefore not be treated as primary standalone indicators for retention prioritisation based on this analysis. They may still provide useful supporting context when combined with stronger segmentation variables.

11.6 Use Financial Exposure Alongside Churn Rate

The analysis demonstrates why churn rate alone is insufficient for financial prioritisation. A segment can contain a smaller number of customers but still represent substantial balance exposure. Combining customer-level churn measures with balance exposure can help organisations distinguish between the scale of customer churn and the potential financial significance of the affected customer population.

11.7 Embed the Findings in Interactive Monitoring

The Streamlit dashboard provides a practical mechanism for exploring the identified segments. Business users can apply filters for geography, age, credit score, tenure, and balance and examine the resulting churn indicators. This approach supports exploratory monitoring and can help users move from broad portfolio-level analysis toward specific customer segments.

Chapter 12. Limitations

Several limitations should be considered when interpreting the findings. These limitations do not invalidate the analysis, but they define the extent to which the results can be generalised or used for causal decision-making.

12.1 Dataset Scope

The analysis is based on 10,000 customer records covering France, Germany, and Spain. The findings therefore describe the observed dataset and should not automatically be generalised to all European banking customers or to other banking institutions.

12.2 Cross-Sectional Nature of the Available Data

The Year field contains only one observed year in the dataset. Consequently, the study cannot establish how churn patterns change over time or whether the observed segment differences represent persistent trends.

12.3 Descriptive Rather Than Causal Analysis

The methodology identifies associations between customer characteristics and churn. It does not establish causality. For example, the higher churn observed among inactive customers does not demonstrate that inactivity directly causes customers to leave. Similarly, demographic differences should be interpreted as observed patterns rather than causal effects.

12.4 Limited Behavioural Information

The available variables provide useful customer-level characteristics, but they do not fully capture behavioural drivers such as transaction frequency, service complaints, digital-banking usage, customer satisfaction, marketing contacts, or product-specific interactions. These factors could provide additional explanatory power in future research.

12.5 Definition of High-Value Customers

High-value customers are operationally defined in this study using the High Balance category, with a balance of 100,000 or above in the dataset's balance units. Actual customer value may depend on profitability, product usage, fees, interest margins, lifetime value, and other measures that are not available in the dataset.

12.6 Financial Exposure Is Not Revenue Loss

The total balance associated with churn is used as a proxy for potential financial exposure. It should not be interpreted as realised lost revenue because the dataset does not provide sufficient information to calculate revenue, profit, fees, interest income, or customer lifetime value.

12.7 Correlation Analysis Limitation

The salary-versus-balance analysis uses Pearson correlation and produces a coefficient of 0.0123. This measures linear association only. A near-zero coefficient does not exclude non-linear relationships, interactions with other variables, or other forms of dependence.

12.8 Segmentation Thresholds

The age, credit-score, tenure, and balance categories were defined using explicit analytical thresholds. Different thresholds could produce different segment sizes and churn rates. The selected categories were chosen for interpretability and alignment with the project methodology rather than being optimised statistically.

12.9 No Predictive Validation

The project focuses on descriptive and comparative churn analytics rather than developing and validating a predictive churn model. Therefore, the results should not be interpreted as individual-level predictions of which customers will churn in the future.

Chapter 13. Conclusion

This study analysed customer churn patterns in a European banking dataset of 10,000 customers using a structured segmentation and comparative analytical framework. The analysis measured overall churn, examined differences across geographic, demographic, engagement, tenure, credit-score, and balance segments, and assessed the financial characteristics of churned customers.

The overall churn rate was 20.37%, with 2,037 customers classified as churned. Germany recorded the highest geographic churn rate at 32.44%, while customers aged 46–60 recorded the highest age-segment churn rate at 51.12%. The combined Germany and 46–60 segment produced the highest observed churn rate at 67.33%. Female customers had a higher churn rate than male customers, at 25.07% compared with 16.46%.

Financial analysis showed that high-balance customers are particularly important from a retention perspective. High-balance customers had a churn rate of 25.23% and represented 59.45% of all churners. Among all churned customers, the total balance associated with churn was 185,588,094.63, of which 159,489,691.00 was associated with high-balance churners. Consequently, high-balance churners accounted for 85.94% of the total balance exposure associated with churn. Engagement also emerged as an important descriptive characteristic. Inactive customers represented 63.92% of churners compared with 44.54% of retained customers. By contrast, credit score and tenure showed comparatively limited variation, and credit-card ownership was broadly similar between retained and churned customers.

Overall, the findings support a segmented approach to churn management. The strongest observed patterns point towards greater analytical attention for high-risk geographic and age combinations, high-balance customers, and customers showing lower engagement. The Streamlit dashboard provides an accessible interface for exploring these patterns and translating the analytical results into a practical decision-support format.

Finally, the study demonstrates the importance of distinguishing between churn rate, churn contribution, and financial exposure. These measures describe different aspects of the churn problem and together provide a more informative view of customer retention risk than an overall churn percentage alone.

Chapter14. References

The references below support the conceptual background, customer retention perspective, churn analytics methodology, banking applications of churn analysis, and the analytical interpretation used in this study. Recent banking-specific studies are included alongside established research on customer defection and churn modelling.

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Chapter 15. Appendix

Appendix A. Dataset Validation Summary

The analytical dataset contained 10,000 customer records and 14 original variables. The validation stage identified no missing values and no duplicate rows. CustomerId was unique across the dataset. The engagement and product fields were checked before analysis.

Validation Item	Result	Interpretation
Dataset size	10,000 rows; 14 columns	Complete analytical dataset
Missing values	0	No missing-value treatment required
Duplicate rows	0	No duplicate records identified
HasCrCard	0 / 1 only	Binary field consistent
IsActiveMember	0 / 1 only	Binary field consistent
Exited	0 / 1 only	Churn label consistent
NumOfProducts	1–4	Valid product-count range

Appendix B. Customer Segmentation Framework

The segmentation framework was created from the analytical requirements of the project. The thresholds were applied consistently across the dataset and were used for descriptive comparison rather than as predictive model thresholds.

Dimension	Segment	Rule
Geography	France / Germany / Spain	Existing categorical field
Age	Under 30	< 30
Age	30–45	30–45
Age	46–60	46–60
Age	60+	60 or above
Credit Score	Low	< 580
Credit Score	Medium	580–669
Credit Score	High	≥ 670
Tenure	New	0–3 years
Tenure	Mid-term	4–7 years
Tenure	Long-term	8–10 years
Balance	Zero Balance	= 0
Balance	Low Balance	> 0 and < 100,000
Balance	High Balance	≥ 100,000

Appendix C. Key Analytical Results

Analytical Indicator	Result
Overall churn rate	20.37%
Highest geography churn rate	Germany: 32.44%
Highest age-segment churn rate	46–60: 51.12%
Female churn rate	25.07%
Male churn rate	16.46%
Highest credit-score segment churn	Low: 22.02%
Highest tenure-segment churn	New: 21.14%
Highest balance-segment churn	High Balance: 25.23%
Germany + 46–60 churn rate	67.33%
High-balance churners	1,211
Total balance exposure associated with churn	185,588,094.63
High-balance share of churned balance exposure	85.94%
Salary $\times$ balance Pearson correlation	$r = 0.0123$

Appendix D. Interpretation Notes

- Churn rates describe observed differences between customer groups and should not be interpreted as causal effects.
- The 63.92% inactive figure represents the composition of churners who were inactive, rather than the churn rate of inactive customers.
- Balance exposure is used as a proxy for potential financial risk associated with churn. It is not a direct measure of lost revenue, profit, fees, or customer lifetime value.
- The salary–balance Pearson correlation of 0.0123 indicates an essentially negligible linear association within the high-balance churner group.
- The high-balance threshold of 100,000 is an analytical segmentation rule used for this project and does not represent a universal banking definition of a high-value customer.

Appendix E. Supporting Analytical Outputs

The project repository contains the supporting charts and consolidated analytical findings used throughout the report. The main report intentionally presents only the visualisations required to communicate the principal findings. Additional charts are retained as supporting project outputs for reproducibility and further exploration.

- Customer distribution by geography, age, credit score, tenure and balance segment.
- Overall and segment-wise churn rate charts.
- Churn contribution charts by geography, age, credit score, tenure and balance.
- Retained versus churned customer profile comparisons.
- Geography $\times$ age churn analysis.
- High-balance churner balance distribution.
- Salary versus balance scatter plot for high-balance churners.

- Balance exposure associated with churn.
- Consolidated analytical findings table.

Appendix F. Dashboard

The Streamlit dashboard developed for this project provides interactive exploration of the same segmentation and churn measures reported in the research paper. Users can filter customers by geography, age segment, credit-score segment, tenure segment and balance segment. The dashboard also presents high-value customer risk and balance-exposure views.

